B. Fedor and New Game

time limit per test 1 second
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1100 | Tags: bitmasks, brute force, constructive algorithms, implementation |
https://codeforces.com/problemset/problem/467/B

After you had helped George and Alex to move in the dorm, they went to help their friend Fedor play a new computer game «Call of Soldiers 3».

The game has (m + 1) players and n types of soldiers in total. Players «Call of Soldiers 3» are numbered form 1 to (m + 1). Types of soldiers are numbered from 0 to n - 1. Each player has an army. Army of the i-th player can be described by non-negative integer x_i. Consider binary representation of x_i: if the j-th bit of number x_i equal to one, then the army of the i-th player has soldiers of the j-th type.

Fedor is the (m + 1)-th player of the game. He assume that two players can become friends if their armies differ in at most k types of soldiers (in other words, binary representations of the corresponding numbers differ in at most k bits). Help Fedor and count how many players can become his friends.

Input

The first line contains three integers n, m, k ($1 \le k \le n \le 20$; $1 \le m \le 1000$).

The i-th of the next (m + 1) lines contains a single integer x_i ($1 \le x_i \le 2^n - 1$), that describes the i-th player's army. We remind you that Fedor is the (m + 1)-th player.

Output

Print a single integer — the number of Fedor's potential friends.

Examples

Input
7 3 1 8 5 111 17

Output
0

Input
3 3 3 1 2 3 4

Output
3